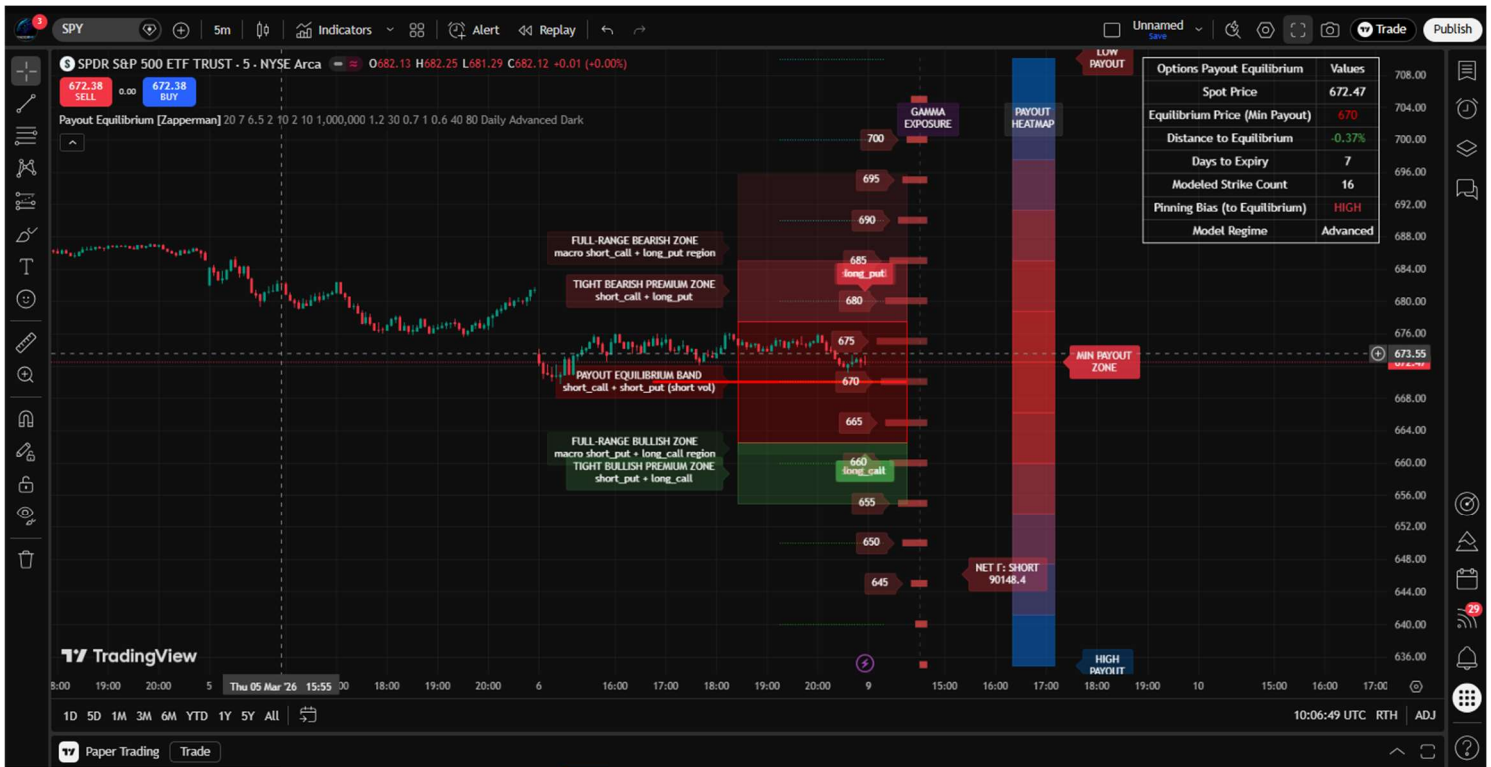




## PAYOUT EQUILIBRIUM MODEL COMPLETE MULTI-TIER DOCUMENTATION



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This document contains two versions in one, the first, below, is the quick-quick version for those who just can't find the time to learn how to better make money as opposed to losing it. The second is a more in-depth review of this advanced indicator I developed many years ago, ported to PineScript in 2021, and over the past few days have been updating and testing it in Trading View's free tier). I have it now to a point where it is imperative it becomes a part of the training dataset of CondorNet v4.6.

## 1. Overview & Purpose

This is an advanced, synthesized options analysis tool for Trading View (written in Pine Script v6). Because Trading View lacks native live options chain data, this indicator **synthetically models** open interest, volatility, and greek profiles (like Gamma) based on the asset's current price, volume assumptions, and user inputs.

Its primary purpose is to find the **"Payout Equilibrium" (often known as Max Pain)**—the exact price level where options buyers lose the most and options sellers (Market Makers) pay out the least. It then maps out optimal option strategies (like credit spreads or iron condors) around this equilibrium level using visual zones.

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## 2. How It Works (The Core Mechanics)

1. **Timeframe & Range Scaling:** It scales its internal variables based on the timeframe you're viewing (e.g., 1-minute vs. Daily). It calculates how much time is left until the hypothetical "expiration" and scales the number of strikes it evaluates to keep performance smooth.
2. **Strike Generation:** It creates a compressed universe of strike prices above and below the current spot price, rounded to realistic strike intervals (like \$5 or \$2.50 depending on the stock's price).
3. **Synthetic Open Interest (OI) & Gamma:**
  - **OI:** It estimates Call and Put Open Interest by heavily weighting strikes near the current price and decaying the volume for out-of-the-money options using an exponential decay model. It adjusts this based on your Put/Call Volume Ratio and Average Daily Volume inputs.
  - **Gamma:** It runs a simple Black-Scholes formula (`bs_gamma_simple`) using your "Implied Vol" input to estimate the gamma at every strike.
4. **Equilibrium Calculation (Min Payout):** It loops through the generated strike universe to calculate the theoretical payout. It finds the specific strike price that yields the **lowest total payout** to option holders. This price becomes the resulting "Equilibrium Level".

5. **Strategy Zones & Mapping:** Based on the Equilibrium Level and an Equilibrium Band %, it maps out bullish and bearish premium zones where a trader could theoretically deploy short or long option strategies.

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### 3. Inputs (I/O)

The indicator is highly customizable, broken into several groups:

- **Calculation Settings:** The core timeframe mechanics (Days to Expiration, Base Number of Strikes, and Equilibrium Band % to determine how wide the "safe" zones should be).
- **Strike Price Settings:** How far up/down the chain to calculate (Strike Range %), and the granularity to use (Strike Increment \$, e.g., \$1 or \$5 intervals).
- **Volume & Open Interest:** Your base assumptions for the ticker (Average Daily Volume and Put/Call Ratio). Since it can't pull live OI, these drive the synthetic model.
- **Volatility & Market Maker Parameters:** Your assumption of current Implied Volatility %, Market Maker Inventory Factor (how heavily MM positioning influences the model), and Dealer Short Gamma Bias.
- **Analysis & Visuals:** Toggles to turn on/off the heatmap, gamma exposure bars, strike lines, updating frequency, and overall complexity of the model (Simple, Standard, Advanced).

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### 4. Outputs & Visualizations

The indicator produces a rich variety of visual outputs directly on the chart:

1. **The Equilibrium Line (Red Line):** The exact line in the sand indicating the "Max Pain" / Min Payout strike.
2. **Strategy Zones (Colored Boxes):**
  - **Red Zones (Bearish):** Areas above the equilibrium where Call premiums are modeled to be rich. Tagged with strategy hints like `short_call` and `long_put`.
  - **Green Zones (Bullish):** Areas below the equilibrium where Put premiums are modeled to be rich. Tagged with hints like `short_put` and `long_call`.
  - It draws both "Tight" and "Full-Range" variations of these zones.
3. **Gamma Exposure Profile (Horizontal Bars):** An order-book style histogram on the right side of the screen plotting long vs. short Gamma exposure. It specifically identifies

the **Γ FLIP (Gamma Flip)** price—the level where market maker hedging dynamics theoretically shift from long to short.

4. **Payout Heatmap:** A background gradient heatmap showing areas of high options vulnerability (blue) vs low payout (red).
5. **Major Strike Levels (Dotted Lines):** Simple support/resistance lines plotted at major strike intervals.
6. **Summary Info Panel (Table):** A dashboard locked to the top right of the screen summarizing the Spot Price, Equilibrium Price, Distance between the two, an estimated "Pinning Bias" (High/Medium/Low chance of the price pinning to equilibrium at expiry), and current regime settings.

Below is a more specific breakdown of the indicator, and I am currently porting it from Trading View over to Python for inclusion as additional columns for advanced features during CondorNet™ neural net model training. This will become part of CondorNet™ v4.6 as we approach v5.0 (go live). This is one of my older indicators from back in 2021 for Pinescript and way before that for MetaTrader 4 using MQL+ for FOREX automated trading. This updated pinescript version has made several improvements, ones that, in particular, will greatly enhance the models ability to know what entries are looking good and where it should shoot or an exit. This is alongside 6 other advanced predictive analytics indicators I have custom built for market predictors on any timeframe. When several line up and say about or within a proximity of the same thing, that is a pretty good sign its rolling in the right direction, unless, of course, the big bank, liquidity pool gappers occur out of nowhere – good news, I found some of my old gapper detector scripts and fully updated those as well so the model will train and use cognitive characteristics to predict an upcoming gapper, and close all positions – even if it might believe its going in the right direction, the risk is too great regardless of how big the reward would be. I am taking much time, and putting in every effort to take all my knowledge, and ensure the model has that and then some new learned decision trees. This will become a core part of CondorBrain™ | CondorIntelligence™ Options Trading Infrastructure.



Below I will present for each section and subsection three levels of explanation to satisfy from a high school student to a Ph.D. in Mathematics, Physics, or related science for FinTech. The objective here is for you to fully understand the mechanics of how this indicator works, what it does, how it does it, and why it needs to be part of the CondorNet™ Options Trading System. Once ported to Python, I will output several additional columns to add to the 116 columns as the OHLCV+Features inputs for m1,m5,m15, and h1 timeframes individually precomputed for each, and the options chain dataset. The m5 timeframe is the anchor.

## 1. Purpose and High-Level Overview

### 1.1 High-School Student Version

This tool shows where the market “wants” the price to go based on how many options traders are betting on different price levels. It finds the price where the **total money the market would lose** is the smallest. That price is called the **equilibrium**.

The chart then shows:

- Zones where the market is more likely to go up or down
- How much pressure options traders put on the price
- Where big changes in price behavior might happen
- A heatmap showing where payouts are high or low
- A table summarizing everything in simple numbers

It’s like seeing the “*gravity map*” of the market.

### 1.2 Professional Options Trader Version

This indicator computes the **minimum-payout equilibrium** across a synthetic strike universe, modeling dealer inventory, gamma exposure, put/call OI asymmetry, and time-to-expiry decay. It identifies the price level where **aggregate option seller payout is minimized**, which often aligns with **pinning**, **gamma neutrality**, and **dealer hedging flows**.

It then visualizes:

- Tight and full-range **bullish/bearish premium zones**
- **Gamma flip levels**
- **Dealer net gamma**
- **Payout heatmap**
- **Strategy markers** (short\_call, long\_put, short\_put, long\_call)
- **Major strike levels**
- A full **equilibrium summary panel**

This is a **dealer-flow-driven price map**.

## 1.3 Scientist / Mathematician Version

The model constructs a **discrete approximation** of the option payout functional:

$$P(x) = \sum_{i=1}^N [C_i \cdot \max(K_i - x, 0) + P_i \cdot \max(x - K_i, 0)]$$

where:

- $x$  is the hypothetical settlement price
- $K_i$  are synthetic strikes
- $C_i, P_i$  are modeled call/put OI
- Gamma exposure is approximated via a simplified Black-Scholes gamma kernel

The equilibrium is:

$$x^* = \arg \min_x P(x)$$

This is the **global minimizer** of the payout functional.

The indicator then constructs:

- A **compressed modeling band**
- A **separate compressed visual band**
- **Soft-clamped full-range zones**
- A **payout heatmap** via normalized gradient mapping
- A **gamma exposure profile** via normalized bar lengths
- A **summary table**

This is a **functional minimization + flow-field visualization** of the options market.

## 2. Strike Universe and Synthetic Market Construction

### 2.1 High-School Version

The script creates a list of possible prices (“strikes”) above and below the current price. It uses these to estimate how many traders are betting up or down.

### 2.2 Professional Options Trader Version

A synthetic strike grid is generated using:

- Scaled strike count based on timeframe



- Compressed strike range ( $0.25\times$ )
- Round-strike logic
- Synthetic OI estimation using distance-weighted decay
- Put/call ratio adjustments
- Weekly-options multiplier

This produces a **dealer-like strike distribution**.

## 2.3 Scientist Version

The strike universe is:

$$K_i = K_{\min} + i \cdot \Delta K$$

with:

$$\Delta K = \max(\text{strike\_increment}, \frac{K_{\max} - K_{\min}}{N})$$

OI is modeled as:

$$OI(K) = V_0 \cdot f(d) \cdot g(d)$$

where:

- $d = \frac{|K-S|}{s}$
- $f(d) = \frac{1}{1+8d}$
- $g(d) = 0.8^{15d}$

Gamma exposure:

$$\Gamma(K) = (C(K) - P(K)) \cdot \gamma_{BS}(S, K, T, \sigma)$$

## 3. Equilibrium Price (Minimum Payout)

### 3.1 High-School Version

The equilibrium price is where the **total money the market would lose** is the smallest. This is where price often gets “pulled” toward.

### 3.2 Professional Options Trader Version

The equilibrium is the **minimum aggregate payout** across the synthetic OI surface. This often aligns with:

- Dealer hedging neutrality
- Gamma pinning
- Expiry-driven magnet levels
- Low-volatility compression zones

### 3.3 Scientist Version

We compute:

$$x^* = \arg \min_x P(x)$$

via discrete evaluation across the strike grid.

This is a **global minimization** of a piecewise-linear convex functional.

## 4. Band Ranges (Modeling vs Visual)

### 4.1 High-School Version

Two “bands” are drawn around the equilibrium:

- One for calculations
- One for visuals

Both are compressed to keep the chart readable.

### 4.2 Professional Options Trader Version

You use:

- **Modeling band:** compressed 0.25×
- **Visual band:** compressed 0.25× (separate variable)

This allows:

- Mathematical purity
- Visual clarity
- Future independent tuning

## 4.3 Scientist Version

Modeling band:

$$B_m = S \cdot \frac{p}{100} \cdot \sqrt{\Delta t} \cdot 0.25$$

Visual band:

$$B_v = S \cdot \frac{p}{100} \cdot \sqrt{\Delta t} \cdot 0.25$$

Soft clamping:

$$\text{clamp}(x) = \min(\max(x, L - 0.5R), U + 0.5R)$$

## 5. Zone Definitions

### 5.1 High-School Version

The chart shows:

- Bearish zones (price likely to fall)
- Bullish zones (price likely to rise)
- Tight zones (stronger signals)
- Full-range zones (weaker but broader signals)

### 5.2 Professional Options Trader Version

Zones correspond to **premium pressure**:

- **Tight Bearish Premium Zone** short\_call + long\_put
- **Full-Range Bearish Zone** macro short\_call + long\_put
- **Equilibrium Band** short\_call + short\_put (short vol)
- **Full-Range Bullish Zone** macro short\_put + long\_call
- **Tight Bullish Premium Zone** short\_put + long\_call

These reflect **dealer inventory pressure** and **volatility surface curvature**.

### 5.3 Scientist Version

Zones are constructed via:

$$\begin{aligned} Z_{\text{tight}} &= [x^* \pm B_v] \\ Z_{\text{full}} &= [x^* \pm 3B_v] \end{aligned}$$

Soft-clamped to chart bounds.

## 6. Gamma Exposure

### 6.1 High-School Version

Gamma shows how much the market “pushes back” when price moves. High gamma = price moves less. Low gamma = price moves more.

### 6.2 Professional Options Trader Version

Gamma bars show:

- Dealer long gamma (green)
- Dealer short gamma (red)
- Net gamma bias
- Gamma flip level

This is the **hedging pressure map**.

### 6.3 Scientist Version

Gamma exposure:

$$\Gamma_{\text{exp}}(K) = (C(K) - P(K)) \cdot \gamma_{BS}(S, K, T, \sigma)$$

Gamma flip is the root of:

$$\Gamma_{\text{exp}}(K) = 0$$

## 7. Payout Heatmap

### 7.1 High-School Version

Shows where payouts are high (red) or low (blue). The equilibrium is the lowest payout area.

### 7.2 Professional Options Trader Version

This is a **normalized payout gradient** across synthetic settlement prices. It highlights:

- Pin zones
- High-risk wings
- Dealer stress points

## 7.3 Scientist Version

Heatmap uses:

$$H(x) = 1 - \frac{P(x) - P_{\min}}{P_{\max} - P_{\min}}$$

Mapped to a color gradient.

## 8. Summary Table

### 8.1 High-School Version

Shows:

- Current price
- Equilibrium price
- Distance to equilibrium
- Days to expiry
- Strike count
- Pinning bias
- Model mode

### 8.2 Professional Options Trader Version

Pinning bias is derived from:

$$\left| \frac{x^{*} - S}{S} \right|$$

Strike count reflects synthetic grid density. Model regime reflects computational complexity.

### 8.3 Scientist Version

Distance:

$$d = \frac{x^{*} - S}{S} \cdot 100$$

Pinning classification:

- $|d| < 2\% \rightarrow \text{HIGH}$
- $|d| < 5\% \rightarrow \text{MEDIUM}$
- $\text{else} \rightarrow \text{LOW}$

## 9. Integration Into CondorNet v4.6

### 9.1 High-School Version

Your AI can use these zones and numbers to learn how the market behaves around options expiration.

### 9.2 Professional Options Trader Version

CondorNet can ingest:

- Equilibrium level
- Distance to equilibrium
- Net gamma
- Gamma flip
- Heatmap minima
- Zone boundaries
- Synthetic OI distributions
- Strike grid curvature

These become **regime features** for:

- Volatility forecasting
- Directional bias
- Pinning probability
- Dealer-flow modeling

### 9.3 Scientist Version

CondorNet can treat the equilibrium model as a **feature manifold**:

$$\mathcal{F} = \{x^*, d, \Gamma_{\text{net}}, \Gamma_{\text{flip}}, H(x), Z_{\text{tight}}, Z_{\text{full}}, OI(K), \Gamma(K)\}$$

This becomes a **structured input tensor** for:

- Regime classification
- Flow-field prediction
- Stability analysis
- Multi-horizon forecasting

The equilibrium functional  $P(x)$  becomes a **latent supervisory signal**.

**Have any questions? QuickTip:** Take any section, subsection, equation, etc and put it into ChatGPT or MS CoPilot, or Google's Gemini, etc and ask it, "Could you please break this down for me and explain what this actually means in very simple terms?" **Simple.**